

EgoEMG: A Multimodal Egocentric Dataset with Bilateral EMG and Vision for Hand Pose Estimation

Anonymous submission

Abstract

Surface electromyography (sEMG) captures muscle activity during hand movement, while egocentric vision observes global hand configuration. These complementary signals are promising for robust hand-pose estimation, but no existing dataset jointly provides synchronized bilateral wrist EMG, egocentric vision, and bimanual pose labels. We present EgoEMG, a multimodal egocentric dataset containing bilateral wristband EMG sampled at 2 kHz, 120 Hz IMU signals, egocentric video, external RGB-D video, and motion-capture-derived wrist and finger joint angles. EgoEMG comprises over 18 hours of recordings from 53 participants performing 60 gesture classes—30 unimanual and 30 bimanual. We establish benchmarks for EMG-to-pose, vision-to-pose, and EMG+vision fusion under shared joint-angle targets and Gesture, User, and Both generalization splits. Across seven visual backbones, residual fusion reduces average error relative to the corresponding vision-only baseline. Together, EgoEMG provides a foundation for studying multimodal hand-pose estimation from muscle activity and visual observations.

Introduction

Hand pose estimation from wearable sensing is a building block for augmented reality, prosthetic control, and physical intelligence (Salter et al. 2024; Atzori and Müller 2014; Zhao et al. 2026). Surface EMG is attractive because it measures muscle activation directly (Farina et al. 2014) and remains usable under self-occlusion, motion blur, and challenging lighting. Recent work has shown that EMG can recover detailed hand articulation (Salter et al. 2024; Verma 2025; Lovecchio et al. 2025). At the same time, egocentric vision has emerged as a powerful modality for hand tracking (Pavlakos et al. 2024; Kwon et al. 2021), but relies on line-of-sight and struggles with fast motion blur and self-occlusion during manipulation. These properties suggest a potential complementarity between the two modalities: EMG provides fine-grained finger articulation even when the hand is occluded, while vision recovers global hand configuration and spatial context. Unlocking this complementarity requires a dataset that synchronizes both modalities with pose labels—yet current EMG pose datasets provide no synchronized egocentric video.

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We address these gaps with *EgoEMG*, a controlled multimodal egocentric dataset for bimanual hand pose estimation. *EgoEMG* contains synchronized bilateral wristband EMG (16 total channels, 8 per wrist at 2 kHz), per-hand IMU, egocentric wide-angle RGB, external RGB-D, and hand motion reconstructed into MANO and UmeTrack parameters (Romero, Tzionas, and Black 2017; Han et al. 2022) with wrist articulation angles, covering 53 participants and 60 gesture classes (30 single-hand, 30 bimanual) over 18 hours of recording.

We provide a benchmark with three tasks—EMG-to-pose, vision-to-pose, and EMG+vision fusion—sharing the same joint-angle prediction target and Gesture, User, and Both generalization splits. On the EMG2Pose dataset, EMGFormer-L reduces User+Stage MAE by 16% relative to the best prior reported result. On *EgoEMG*, EMGFormer-M achieves the lowest overall MAE among all evaluated EMG baselines. On vision-to-pose, ResNet backbones outperform ViT at lower parameter counts, while the hand-specialized WiLoR baseline achieves the strongest standalone visual accuracy. Across all seven visual backbones, residual fusion lowers average error relative to the matched vision-only baseline, and the gains become more modest for the strongest visual models.

Our contributions are:

- *EgoEMG*, the first dataset jointly providing bilateral wristband EMG and IMU, egocentric RGB, external RGB-D, bimanual MANO/UmeTrack annotations with wrist joint angles, and per-frame gesture class labels.
- A three-task hand pose estimation benchmark (EMG-to-pose, vision-to-pose, EMG+vision fusion) with Gesture, User, and Both generalization splits under a unified joint-angle prediction target.
- Baseline results showing that EMGFormer provides a strong EMG-to-pose reference on both EMG2Pose and *EgoEMG*, that hand-specialized vision modeling improves over generic ResNet/ViT baselines, and that residual EMG+vision fusion improves accuracy across the evaluated visual backbones.

Related Work

Hand pose datasets and benchmarks. Public EMG datasets cover gesture recognition (Atzori and Müller 2014; Pradhan, He, and Jiang 2022; Kaczmarek, Mańkowski, and Tomczyński 2019; Du et al. 2017; Amma et al. 2015), high-density recordings (Jiang et al. 2021), force estimation (Seo

et al. 2024), typing (Sivakumar et al. 2024), and demographically diverse cohorts (Gowda et al. 2025). Recent work also combines wrist-worn sEMG and IMU with egocentric video to obtain force-enriched manipulation demonstrations (He et al. 2026). For continuous pose estimation, EMG2Pose (Salter et al. 2024) provides synchronized bilateral EMG with mocap-derived joint angles, while MyoKi (Andreas et al. 2025) and Jarque-Bou et al. (2019) pair forearm EMG with instrumented-glove kinematics. Cross-user generalization has been studied through benchmarks such as EMGBench (Yang et al. 2024). In contrast, visual hand-pose benchmarks provide rich RGB annotations for monocular, egocentric, and hand-object settings (Zimmermann et al. 2019; Yu et al. 2022; Moon et al. 2023; Banerjee et al. 2024; Kwon et al. 2021; Huang et al. 2024), but do not include synchronized wrist EMG. Existing resources therefore lack a unified benchmark that combines egocentric video, bilateral wristband EMG, wrist articulation, and continuous hand-pose labels, limiting the study of multimodal pose regression from both visual observations and muscle activity.

EMG-based pose estimation. Decoding hand kinematics from surface EMG has progressed from single-DoF (Ngeo, Tamei, and Shibata 2012, 2014) and multi-DoF finger-angle regression to full-hand pose estimation. NeuroPose (Liu, Zhang, and Gowda 2021) demonstrated 3D hand-pose tracking from wearable EMG, while EMG2Pose (Salter et al. 2024) later established a larger-scale benchmark using a TDS+LSTM architecture. Recent work explores tendon-driven motion modeling (Verma 2025), vector-quantized pose tokenization (Lovecchio et al. 2025), short-calibration user adaptation (Xie and Cheung 2026), the role of position versus velocity decoding under the original benchmark protocol (Hadidi et al. 2026), and generalization-focused teleoperation (Zhao et al. 2026). In parallel, EMG foundation models and self-supervised approaches (Fasulo et al. 2025; Mehlman et al. 2025; Raghu, MacIsaac, and Scheme 2024; Laamerad et al. 2024) explore large-scale pretraining, but primarily target gesture classification rather than continuous pose. Despite these advances, EMG-only pose estimation remains sensitive to inter-user variability, and datasets based on instrumented gloves (Andreas et al. 2025; Jarque-Bou et al. 2019) provide limited degrees of freedom and can suffer from sensor drift and hysteresis.

Vision-based hand pose estimation and multimodal fusion. Monocular RGB hand-pose estimation has advanced rapidly with the MANO hand model (Romero, Tzionas, and Black 2017), large-scale annotated datasets (Zimmermann et al. 2019; Yu et al. 2022; Moon et al. 2023; Huang et al. 2024), and Transformer-based reconstructors (Pavlakos et al. 2024; Potamias et al. 2024). Egocentric systems and datasets such as UmeTrack (Han et al. 2022), HOT3D (Banerjee et al. 2024), H2O (Kwon et al. 2021), and WristPP (Xi et al. 2026) further support first-person hand tracking and interaction understanding in wearable and egocentric settings. However, vision-based methods remain fundamentally constrained by line of sight: self-occlusion, inter-hand occlusion, object manipulation, and motion blur can substantially degrade pose estimates. EMG provides complementary muscle-activation

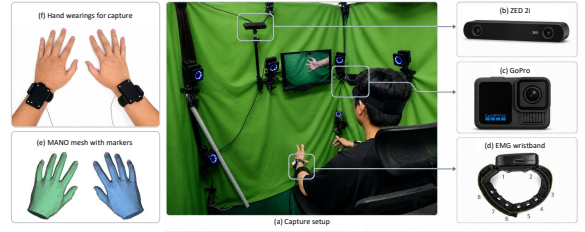


Figure 1: Data collection setup for *EgoEMG*. Bilateral EMG wristbands, head-mounted egocentric RGB, external ZED 2i RGB-D, and optical motion capture with hand markers for pose labels.

cues that are independent of image visibility. Recent work has begun exploring EMG+vision multimodal learning for gesture understanding (Hao et al. 2024; Zandigohar et al. 2024; Cui et al. 2026) and cross-modal EMG-pose representation (Cui et al. 2025; Liu et al. 2025), but these efforts focus on gesture classification or offline representation mapping rather than continuous hand-pose regression. The lack of synchronized egocentric video, bilateral EMG, and continuous pose labels has therefore prevented systematic study of EMG+vision fusion for hand-pose regression.

EgoEMG Dataset

Overview and Capture Setup

EgoEMG is collected with a synchronized multimodal capture setup, including bilateral EMG, IMU, egocentric RGB, external RGB-D, and optical motion capture. Each participant wears two WAVELETECH EMG wristbands, one on each wrist, with each wristband recording 8 channels of surface EMG at 2 kHz. The visual setup contains a head-mounted wide-angle RGB camera for egocentric hand observation and an external ZED 2i RGB-D camera mounted on the motion-capture rack. Hand motion is tracked using 21 reflective markers per hand at 120 Hz with an FZMotion optical motion capture system. All modalities are temporally synchronized via host-timestamp soft-synchronization followed by linear interpolation to a common timeline (Figure 1).

We recruited 53 participants. *EgoEMG* contains 60 gesture classes (30 single-hand, 30 bimanual) with per-frame gesture labels for gesture-conditioned evaluation. Gesture prompts are displayed in randomized order; single-hand gestures are performed mirror-symmetrically with both hands while varying wrist orientation; bimanual gestures follow demonstrated two-hand interactions, capturing natural coordination patterns. Figure 2 shows representative synchronized samples from all modalities.

Pose Label Reconstruction

Hand pose labels are reconstructed from optical motion-capture markers into MANO parameters (Romero, Tzionas, and Black 2017). We use a learning-based *markers2mano* pipeline that maps 21 marker positions to MANO pose and shape parameters. The pipeline uses fixed MANO surface

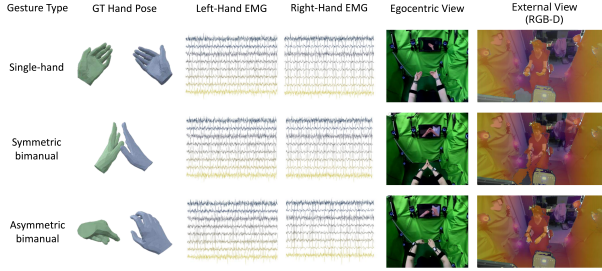


Figure 2: Representative synchronized samples from *EgoEMG*. Each row (6 s window) shows GT hand pose, bilateral EMG, egocentric RGB at window center, and external RGB-D.

vertices as virtual marker correspondences, allowing captured 3D marker positions to be aligned with the canonical MANO mesh. Full details of the *markers2mano* architecture, training data, and augmentation pipeline are provided in the supplementary material.

Compared with the per-frame inverse-kinematics solver used by EMG2Pose, which has a reported 12.7% invalid-frame rate, our learning-based reconstruction reduces the invalid-frame rate to 3.6%, a $3.5\times$ reduction. The resulting MANO meshes achieve a mean marker-to-mesh alignment error of 4.3 mm. In addition to MANO parameters, *EgoEMG* provides per-frame joint angles in a 22-DoF scalar format (20 finger angles plus 2 wrist articulation angles) obtained by optimizing UmeTrack joint angles to match MANO landmark positions; the conversion procedure is described in the supplementary material. As an independent cross-modal check, we reproject reconstructed MANO meshes onto egocentric RGB frames using the calibrated camera model. Visual inspection of randomly sampled reprojections confirms consistent mesh-to-image alignment across gesture families (supplementary material).

Wrist articulation angles are computed directly from motion-capture markers, independent of the MANO reconstruction pipeline. Markers attached to each EMG armband define a forearm coordinate frame, from which we compute two wrist degrees of freedom: flexion/extension and radial/ulnar deviation; the full derivation is provided in the supplementary material.

Dataset Statistics and Positioning

Table 1 compares *EgoEMG* with representative public EMG datasets. Among existing datasets, only EMG2Pose and *EgoEMG* provide bilateral EMG with motion-capture-derived continuous hand-pose labels. *EgoEMG* further contributes synchronized egocentric RGB, external RGB-D, IMU, and explicit wrist articulation labels, enabling direct study of cross-modal hand pose estimation.

Hand shape and gesture diversity. Figure 3(a) shows the MANO shape distribution across 53 participants, demonstrating substantial inter-subject hand-shape variation. The 60 gesture classes include 30 single-hand gestures and 30 bimanual interaction gestures, covering both symmetric co-

Table 1: Comparison of *EgoEMG* with representative public EMG datasets. “EMG Ch.” counts total channels across both wrists (per-wrist $\times 2$ for bilateral datasets); “Bilateral” indicates dual-wrist EMG capture; “Wrist labels” indicates explicit wrist articulation labels; “Pose labels” indicates continuous hand pose annotations.

Dataset	Subj.	EMG Ch.	Bilateral	Ego RGB	RGB-D	Wrist labels	Pose labels	Gestures
Jarque-Bou et al. (2019)	22	8	—	—	—	—	18 (g)	26
NinaPro (Azori and Müller 2014)	27-78	10-16	—	—	—	✓	22 (g) [*]	53
MyoKi (Andreas et al. 2025)	36	16	—	—	—	—	18 (g)	74
PiMForce (Seo et al. 2024)	21	8	—	—	—	—	20 (g)	63
EMG2Pose (Salter et al. 2024)	193	32 (16 $\times 2$)	✓	—	—	—	20 (m)	50
<i>EgoEMG</i> (ours)	53	16 (8$\times 2$)	✓	✓	✓	✓	22 (m)	60

^{*} NinaPro DB1, 2 and 5 provide continuous joint angles (22-DoF); remaining sub-databases provide gesture labels only. (g) = instrumented glove; (m) = optical motion capture.

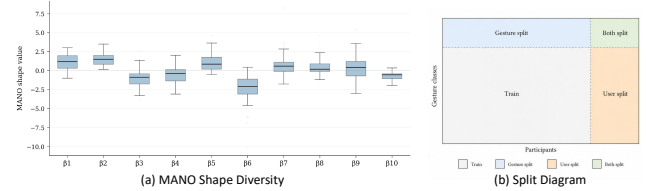


Figure 3: (a) MANO shape diversity across *EgoEMG* participants (106 hands). (b) Evaluation splits on the participant-gesture matrix (Train, Gesture/User/Both).

ordination and asymmetric manipulation. The full gesture vocabulary is provided in the supplementary material.

Evaluation Splits and Release Format

Following the protocol of EMG2Pose (Salter et al. 2024), we define two primary generalization axes: *gesture* and *user*. Based on these axes, we evaluate three mutually exclusive splits: a *Gesture split*, where held-out gestures are evaluated on seen participants; a *User split*, where held-out participants are evaluated on seen gestures; and a combined *Both split*, where both gestures and participants are held out. All modalities follow the same split assignment, preserving synchronized multimodal examples and preventing leakage along the held-out axis. We hold out 10 gestures for the gesture split and 6 participants for the user split, with an overall train-to-validation/test ratio of approximately 7:3. Figure 3(b) illustrates the split design.

The released dataset includes synchronized EMG and IMU signals, egocentric RGB frames, external RGB-D frames, MANO-based and UmeTrack-based hand-pose annotations, wrist articulation angles, timestamps, and calibration metadata. The prediction target is a 22-DoF joint-angle vector per hand: 20 finger angles and 2 wrist articulation angles. The same target definition and semantic ordering are used for both left and right hands.

Benchmark and Baselines

Benchmark Tasks

As an initial benchmark, *EgoEMG* defines three prediction tasks under the same 22-DoF target definition and evaluation split axes. The target consists of 20 finger joint angles and 2 wrist articulation angles per hand. The three tasks are: *EMG-to-pose*, which predicts joint angles from EMG

windows; *vision-to-pose*, which predicts joint angles from single egocentric RGB hand crops; and *EMG+vision fusion*, which uses both modalities to predict the same output. The external RGB-D and IMU streams are included in the release to support future extensions.

Evaluation Protocol

We evaluate all tasks on the Gesture, User, and Both splits defined in the dataset section above. The primary metric is mean absolute error (MAE) over joint angles:

$$\text{MAE} = \frac{1}{JT} \sum_{j=1}^J \sum_{t=1}^T |\hat{\theta}_{j,t} - \theta_{j,t}| \quad [^\circ], \quad (1)$$

where J is the number of predicted joint angles and T is the number of evaluated time steps. For wrist analysis, we additionally report MAE on flexion/extension and radial/ulnar deviation. Unless otherwise specified, all errors are reported on the held-out test split. For vision-only and fusion evaluation, we use 256×256 hand crops localized by projecting motion-capture keypoints into the calibrated egocentric camera. These models predict the window center frame; fusion pairs that image with a centered, non-causal 6 s EMG window. The resulting protocol evaluates offline pose regression given hand localization rather than hand detection or causal tracking. EMG-only evaluation instead uses the full predicted trajectory over each window.

EMG Baseline

As the EMG-to-pose baseline, we introduce EMGFormer (Figure 4), a temporal model that maps filtered EMG windows to per-frame joint angles. For the reported *EgoEMG* experiments, each example contains 12,000 samples (6 s) from the target wrist’s eight EMG channels. A TDS-style temporal convolutional frontend (Hannun et al. 2019) downsamples the 2 kHz signal to latent features at approximately 38 Hz, followed by a Transformer decoder with rotary positional encoding (RoPE) (Su et al. 2024) to model long-range temporal dependencies. Left-hand EMG is mirrored to the right-hand convention during training. Each forward pass predicts one hand’s joint-angle trajectory, including wrist articulation for *EgoEMG*. We provide three model sizes; implementation details, architecture specifications, and ablations are provided in the supplementary material.

Vision Baseline

For vision-to-pose, we evaluate seven single-frame reference baselines. Six are generic image backbones across two families: ResNet (He et al. 2016) ({18, 50, 152}) and ViT (Dosovitskiy et al. 2021) ({Small, Base, Large}). ResNet backbones are initialized from ImageNet (Deng et al. 2009)-pretrained weights, while ViT backbones are initialized from DINOv2 (Oquab et al. 2024)-pretrained weights. Each generic model takes a 256×256 egocentric hand crop from the window center frame, extracts a feature vector with the backbone, and regresses the 22-DoF target through an MLP prediction head. We additionally evaluate WiLoR (Potamias et al. 2024) as a hand-specialized vision baseline initialized from a pre-trained MANO-based hand-pose checkpoint and fine-tuned

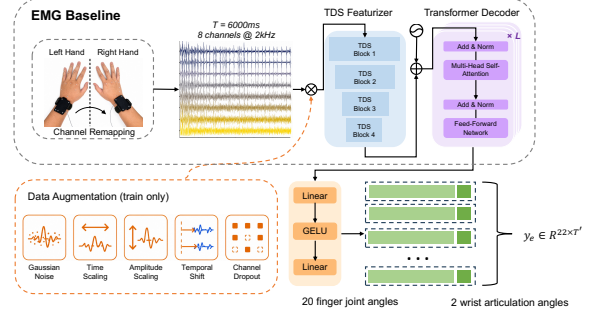


Figure 4: EMGFormer architecture for EMG-to-pose. Target-wrist EMG is encoded by a TDS-style temporal convolutional frontend, decoded by a Transformer with RoPE positional encoding, and projected to joint angles including wrist articulation.

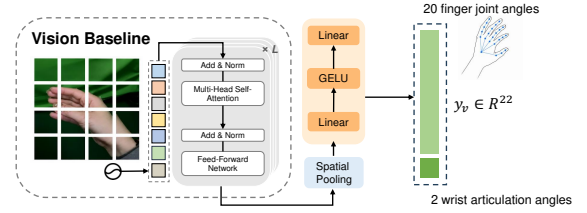


Figure 5: Vision-to-pose baseline. A 256×256 hand crop from the center frame is processed by a ResNet or ViT backbone, followed by an MLP head that regresses 22 joint angles.

on the same protocol (Figure 5). Model sizes and training configurations are provided in the supplementary material.

Fusion Baseline

The EMG+vision baseline is designed as a simple residual fusion model (Figure 6). The vision branch first predicts a vision-only pose $\mathbf{y}_v$, while the EMG branch predicts an additive correction $\Delta \mathbf{y}_{\text{emg}}$. The final prediction is $\mathbf{y} = \mathbf{y}_v + \Delta \mathbf{y}_{\text{emg}}$. This design makes the fusion model start from a strong visual baseline and learn the incremental pose information contributed by EMG.

The vision branch extracts a 256-d feature from the center-frame hand crop and uses a prediction head to produce $\mathbf{y}_v \in \mathbb{R}^{22}$. The EMG branch uses EMGFormer to encode the full EMG window into a temporal feature sequence, followed by learned temporal attention pooling to obtain a 256-d EMG feature. The two features are concatenated and passed to a fusion head that predicts $\Delta \mathbf{y}_{\text{emg}} \in \mathbb{R}^{22}$. The last layer of the residual head is zero-initialized so that the model begins from the vision-only prediction. We train the fusion model under a center-supervised protocol: only the window center frame is supervised, matching the vision-only target and enabling direct comparison.

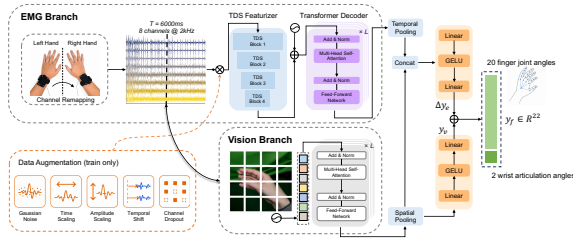


Figure 6: EMG+vision fusion baseline. Vision branch predicts y_v ; EMG branch predicts residual Δy_{emg} (zero-initialized head).

Table 2: EMG-to-pose results on *EgoEMG*. All baselines are retrained under the same evaluation protocol. MAE is reported in degrees; lower is better. Gesture/User/Both columns report per-user mean $\pm$ standard deviation. Avg is the per-sample weighted mean across all test splits. The shaded row marks the best accuracy–capacity trade-off; bold marks the lowest unambiguous result in each column.

Method	Params	Gesture	User	Both	Avg.
<i>EMGFormer variants</i>					
EMGFormer-S	3.5M	12.3 $\pm$ 1.5	16.0 $\pm$ 0.6	16.3 $\pm$ 1.5	14.1
EMGFormer-M	6.6M	11.7$\pm$1.6	15.9 $\pm$ 0.6	16.4 $\pm$ 1.5	13.8
EMGFormer-L	16.3M	11.9 $\pm$ 1.6	16.0 $\pm$ 0.8	16.4 $\pm$ 1.2	13.9
<i>Reimplemented baselines</i>					
emg2pose (Salter et al. 2024)	3.0M	15.5 $\pm$ 1.3	14.8$\pm$2.9	16.3 $\pm$ 0.6	15.8
vemg2pose (Salter et al. 2024)	6.0M	15.0 $\pm$ 1.4	16.3 $\pm$ 1.7	17.3 $\pm$ 1.3	15.9
NeuroPose (Liu, Zhang, and Gowda 2021)	6.4M	15.8 $\pm$ 1.2	15.7 $\pm$ 1.3	16.3 $\pm$ 0.7	16.1
SensingDynamics (Simpetru et al. 2024)	1.0M	16.2 $\pm$ 1.1	16.4 $\pm$ 0.3	16.7 $\pm$ 0.8	16.4

Experiments

EMG Baseline Results

Table 2 reports EMG-to-pose results on *EgoEMG* for EMGFormer variants and four traditional architectures across the Gesture, User, and Both test conditions. The baselines are the original emg2pose TDS+MLP model and the recurrent vemg2pose model (Salter et al. 2024), NeuroPose, a convolutional encoder-decoder (Liu, Zhang, and Gowda 2021), and SensingDynamics, a convolutional model operating on raw and low-pass EMG (Simpetru et al. 2024).

EMGFormer variants achieve the best accuracy on the Gesture split, indicating that EMGFormer better captures gesture-level articulation under the seen-user, held-out-gesture setting. The original emg2pose baseline retains the lowest User-split error, underscoring the challenge of cross-user generalization, while EMGFormer-S matches the lowest reported Both-split error. EMGFormer-M achieves the best overall MAE and offers the best accuracy–capacity trade-off among the EMGFormer variants, outperforming EMGFormer-L with fewer than half the parameters.

On EMG2Pose (Table 3), EMGFormer-L achieves 12.25 $^{\circ}$ $\pm$ 1.08 $^{\circ}$ MAE on User+Stage, a 16% reduction relative to the best prior reported result of 14.6 $^{\circ}$ $\pm$ 1.3 $^{\circ}$ from Position/Velocity MT (Hadidi et al. 2026). The compact EMGFormer-S reaches 12.34 $^{\circ}$ $\pm$ 1.07 $^{\circ}$ with only 58% of the vemg2pose model size.

Table 3: EMG-only regression results on EMG2Pose test splits. MAE is reported in degrees; lower is better. EMGFormer values report per-user mean $\pm$ standard deviation across users. † indicates numbers reported in the corresponding papers. MT denotes joint Regression+Tracking training, evaluated here in the Regression setting without test-time user calibration. Bold marks the lowest result in each column.

Method	Params	User	Stage	User+Stage
<i>EMGFormer variants</i>				
EMGFormer-S	3.5M	12.5 $\pm$ 1.1	11.1 $\pm$ 1.2	12.3 $\pm$ 1.1
EMGFormer-M	6.6M	12.4 $\pm$ 1.1	10.2 $\pm$ 1.1	12.4 $\pm$ 1.1
EMGFormer-L	16.3M	12.3 $\pm$ 1.1	9.3$\pm$1.1	12.3$\pm$1.1
<i>Prior work (reported)</i>				
emg2pose (position) † (Salter et al. 2024)	3.0M	12.6 $\pm$ 1.3	15.2 $\pm$ 1.6	15.6 $\pm$ 1.3
vemg2pose † (Salter et al. 2024)	6.0M	12.2 $\pm$ 1.3	15.2 $\pm$ 1.6	15.8 $\pm$ 1.4
NeuroPose † (Liu, Zhang, and Gowda 2021)	6.4M	13.2 $\pm$ 1.1	17.2 $\pm$ 1.7	17.5 $\pm$ 1.5
SensingDynamics † (Simpetru et al. 2024)	1.0M	15.5 $\pm$ 1.4	18.8 $\pm$ 1.6	18.7 $\pm$ 1.6
Position MT † (Hadidi et al. 2026)	6.0M	11.5 $\pm$ 1.2	14.0 $\pm$ 1.6	14.6 $\pm$ 1.3
Velocity MT † (Hadidi et al. 2026)	6.0M	11.6 $\pm$ 1.3	13.9 $\pm$ 1.6	14.6 $\pm$ 1.3
CLDM † (Verma 2025)	7.0M	11.3$\pm$1.0	14.3 $\pm$ 1.5	14.7 $\pm$ 1.4

Vision-to-Pose Results

Table 4 reports vision-only and EMG+vision results on *EgoEMG*. We first analyze the vision-only rows, which include six generic single-frame baselines spanning two backbone families, ResNet and ViT, together with one hand-specialized baseline, WiLoR. Among the generic backbones, ResNet-152 achieves the best vision-only performance across all splits. ResNet-50 also outperforms ViT-Large despite using substantially fewer parameters, suggesting that convolutional backbones remain highly competitive as generic egocentric references in this data regime. Scaling reduces error within both generic backbone families, while the User and Both splits remain harder than the Gesture split across all vision-only models. The relative underperformance of generic ViT regressors may reflect the scale of the available supervised training data, as transformer-based vision models often benefit from larger datasets. In contrast, the hand-specialized WiLoR baseline achieves the strongest standalone visual accuracy, reducing the average MAE to 4.7 $^{\circ}$. This indicates that task-specific hand-pose pretraining remains important in the *EgoEMG* setting.

EMG+Vision Fusion Results

We evaluate EMG+vision fusion using the residual architecture described above, with EMGFormer-S as the EMG backbone. The fusion model predicts the window center frame, where the egocentric RGB observation is available, making its MAE directly comparable to the vision-only baseline under the same supervision target. For each visual backbone, we train two fusion variants: one freezes the pretrained visual predictor and the other fine-tunes it jointly with the EMG branch. For each backbone, we retain the frozen or fine-tuned variant with the lower validation MAE. Table 4 reports the retained variant; the complete frozen-versus-fine-tuned comparison is deferred to the supplementary material.

Fusion lowers the sample-weighted average error for all seven backbones, with the largest gains for ResNet-18 and ViT-S and smaller gains for stronger visual models. The

Table 4: Vision-only versus EMG+vision fusion on *EgoEMG*. Each split reports MAE in degrees as Vision → Fusion on identical center frames; bold marks the lower error. Frz./FT denotes a frozen/fine-tuned visual predictor. Params reports total parameters in millions as vision-only/fusion. Avg is the mean and Δ_{avg} is the overall fusion gain computed from unrounded values.

Backbone	Params (V/F)	Update	Gesture	User	Both	Avg	Δ_{avg}
ResNet-18	11.5/15.5M	FT	5.10 → 4.60	6.80 → 6.40	6.80 → 6.40	5.84 → 5.40	+0.44
ResNet-50	23.5/29.0M	Frz.	4.50 → 4.39	6.23 → 6.19	6.10 → 6.05	5.27 → 5.19	+0.09
ResNet-152	58.2/63.6M	Frz.	4.31 → 4.25	6.07 → 6.04	6.03 → 6.03	5.11 → 5.06	+0.05
ViT-S/14	21.9/25.8M	FT	5.00 → 4.50	7.30 → 6.80	7.20 → 6.80	6.02 → 5.54	+0.48
ViT-B/14	86.2/90.2M	Frz.	4.85 → 4.80	7.02 → 7.03	6.86 → 6.84	5.78 → 5.75	+0.03
ViT-L/14	303.8/308.0M	Frz.	4.39 → 4.34	6.67 → 6.69	6.60 → 6.60	5.39 → 5.36	+0.03
WiLoR	631.6/635.8M	Frz.	3.98 → 3.96	5.57 → 5.52	5.58 → 5.54	4.73 → 4.68	+0.04

smaller gains with larger visual backbones may reflect the growing parameter imbalance between the visual backbone and the much smaller EMG and fusion branches, which may make joint multimodal optimization more difficult. The improvements are not uniform across splits: frozen ViT-B and ViT-L improve on unseen gestures but are nearly unchanged or slightly worse on unseen users, whereas WiLoR yields small gains on all three splits. The ablation in the supplementary material indicates that fine-tuning works better for ResNet-18 and ViT-S, while freezing is preferable for larger backbones under our settings. The per-gesture analysis in the supplementary material further examines where fusion is most beneficial.

Effect of self-occlusion. We analyze the matched Vision-RN18 and Fusion-RN18+S models. To quantify self-occlusion, we rasterize each ground-truth MANO mesh in camera coordinates at the native image resolution using the calibrated camera and a z-buffer. A vertex is visible when its depth agrees with the local z-buffer within 5 mm. With w_i denoting the surface-area weight of vertex i , we define

$$s_{\text{occ}} = 1 - \frac{\sum_{i \in \mathcal{V}_{\text{vis}}} w_i}{\sum_{i=1}^V w_i}, \quad (2)$$

where $\mathcal{V}_{\text{vis}}$ is the visible-vertex set; $s_{\text{occ}} = 0$ denotes a fully visible mesh (full details in the supplementary material). Fusion improves on vision-only in every occlusion bin: the mean gain rises from 0.24° in the least-occluded bin to 0.52° in the most-occluded bin, and reaches approximately 0.8° at moderate-to-high occlusion. Figure 7 visualizes this pattern. Although the frame-level association between occlusion and fusion gain is weak, the positive gains across bins are consistent with EMG providing complementary cues when visual evidence is degraded. Figure 8 illustrates a representative case: when visual evidence is ambiguous, EMG retains pose cues independent of image appearance, and the fused model recovers articulation closer to the ground truth.

Discussion and Conclusion

EgoEMG provides a multimodal benchmark for hand pose estimation with synchronized bilateral EMG, IMU, egocentric RGB, external RGB-D, and hand-pose annotations including wrist articulation. By using a shared 22-DoF prediction target

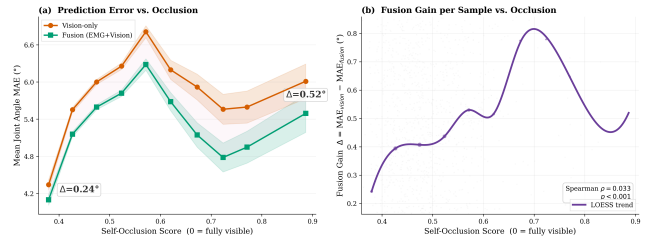


Figure 7: Fusion benefit as a function of hand self-occlusion on the *EgoEMG* test set (16,650 valid predictions). (a) Mean joint-angle MAE of vision-only and EMG+vision fusion across fixed-width (0.05) occlusion bins; annotations show the fusion gain at the lowest and highest bins. (b) Per-bin fusion gain ($\Delta = \text{MAE}_{\text{vision}} - \text{MAE}_{\text{fusion}}$) with a LOESS trend line. Error bands denote ± 1 SEM.

and unified split axes, the benchmark aligns EMG-to-pose, vision-to-pose, and EMG+vision fusion under a common label space while retaining modality-specific supervision protocols.

Our experiments highlight three findings. First, cross-user generalization remains the central challenge for EMG-based pose estimation: errors increase more on unseen participants than on unseen gestures. Second, EMG and vision offer complementary information. Vision provides the stronger standalone signal in the current benchmark, whereas EMG captures pose-relevant cues for fine articulation and improves average accuracy when fused with the evaluated vision baselines. Third, scaling behavior depends on the data regime. The proposed EMGFormer architecture improves over vemg2pose on the original EMG2Pose benchmark, but larger variants do not consistently improve on *EgoEMG*. This suggests that increasing model capacity alone is insufficient under limited subject coverage and motivates large-scale EMG pretraining, cross-dataset transfer, and subject-robust adaptation.

Limitations and future work. The present study has several limitations. First, *EgoEMG* includes 53 participants, which is substantial for synchronized multimodal capture but still limited for modeling the full extent of inter-subject EMG variability. Expanding the dataset to more diverse participants would enable a more systematic study of cross-user

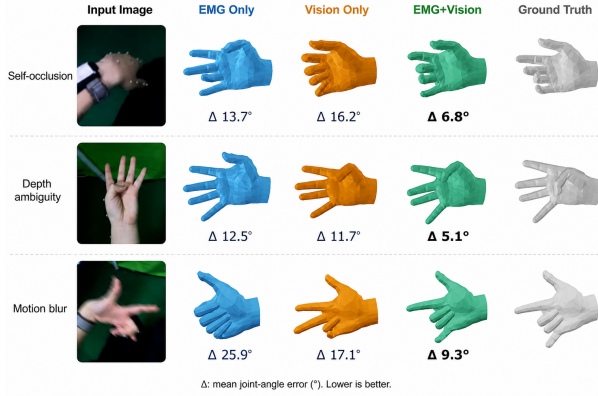


Figure 8: Qualitative comparison on *EgoEMG*. The fused model (green) recovers articulation closer to GT under motion blur, self-occlusion, and depth ambiguity.

robustness. Second, the current vision benchmark is limited to single-frame models. Since *EgoEMG* provides synchronized multi-view video and RGB-D streams, it can support future work on temporal, multi-view, and RGB-D-based methods. The benchmark also does not yet include rich hand-object interaction with instrumented or densely annotated objects; future extensions should test whether the same multimodal signals remain helpful when object contact introduces additional occlusion and pose ambiguity. Third, our fusion baseline is intentionally simple, and its gains are modest for the larger visual backbones, possibly because the growing parameter imbalance between the visual backbone and the compact EMG and fusion branches makes multimodal optimization more difficult. Future work should explore more balanced and expressive fusion architectures, such as cross-attention and uncertainty-aware fusion.

More broadly, *EgoEMG* provides a testbed for studying when physiological signals add information beyond visual observations, especially under unreliable visual evidence such as self-occlusion and fast motion. We hope it will support future work on subject-robust, multimodal hand-pose estimation under a unified benchmark.

Ethics Statement

All participants provided informed consent under an IRB-approved protocol and were compensated at a standard hourly rate commensurate with local guidelines. Participants were explicitly informed of their right to withdraw from the study at any time and to request deletion of their data; no participant exercised this right. The consent form explained the purpose of data collection, the modalities recorded, and the intended research use.

Direct identifiers are removed from the dataset: facial regions in egocentric and external camera frames are automatically blurred, and participant identifiers are replaced with anonymous numeric IDs. While EMG and motion capture data do not reveal sensitive health information beyond gross motor function, we note that MANO shape parameters and EMG signals could in principle serve as soft biometric iden-

tifiers under certain conditions. The dataset will be released for noncommercial research under CC-BY-NC 4.0, accompanied by explicit guidance against re-identification attempts. We commit to hosting the dataset on a stable platform (e.g., Zenodo) with a DOI and versioned releases, and will provide a contact email for data-related inquiries and error reporting.

Reproducibility Statement

The anonymized Code and Data Supplement submitted with this manuscript contains training and evaluation code, exact configuration files, normalization statistics, and a runnable numeric reviewer sample. The supplementary material provides the architecture and optimization settings used for the reported *EgoEMG* experiments. The public release will include dataset documentation, split definitions, preprocessing scripts, benchmark configurations, and an anonymized download link. The reported *EgoEMG* EMGFormer variants use 12,000-sample (6 s) target-wrist EMG windows, are trained for 150 epochs with bf16 mixed precision, and use four to six NVIDIA RTX 4090 GPUs depending on model size.

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